mtDNA-FM: A Domain-Specialized Foundation Model for Mitochondrial DNA with Circular Positional Encoding and Heteroplasmy Modeling

Thawfeek Varusai[1]

[1]Independent Researcher

Correspondence: vthawfeek@gmail.com

Abstract

Mitochondrial DNA (mtDNA) is a 16,569 bp circular genome with central roles in human disease, population genetics, and evolutionary biology. Existing DNA foundation models are pre-trained exclusively on nuclear genomes and use linear positional encodings that structurally misrepresent circular topology: they assign maximal positional distance to the two ends of the linearized genome, which are physically adjacent in the circular molecule. We present **mtDNA-FM**, the first foundation model dedicated to mitochondrial DNA. mtDNA-FM introduces two architectural novelties: (i) a *circular positional encoding* implemented as a fixed sinusoidal buffer such that positions 0 and 16,568 receive identical encodings, eliminating the artificial discontinuity at the genome junction; and (ii) a *heteroplasmy projection channel* that accepts a continuous per-position variant allele fraction (0–1) alongside sequence content. Pre-trained via a two-phase curriculum on 152,590 complete vertebrate mtDNA sequences (Phase 1: 117,615 cross-species; Phase 2: 34,975 human HmtDB sequences), mtDNA-FM produces embeddings with strong zero-shot capabilities without any task-specific labels. Zero-shot 5-nearest-neighbour search on a 26-class haplogroup panel achieves 37.9% accuracy (3.85% random baseline, 9.9× lift; 95% CI 34.4–41.2%), and zero-shot variant embeddings discriminate ClinVar pathogenic mtDNA SNPs from gnomAD common variants with AUROC 0.777 (95% CI 0.731–0.821). The model correctly places archaic human sequences (Neanderthal NC_011137.1, Denisovan FR695060.1) outside modern human variation, and shows gene structure in the embedding space, with rRNA genes separated from protein-coding and tRNA genes, without any functional labels. LoRA fine-tuning for haplogroup classification achieves 1.83% under CPU compute constraints (class collapse documented as a known limitation requiring GPU resources); this reflects a compute limitation, not model capacity, as the zero-shot representations are informative. Pre-trained weights, LoRA adapters, a Gradio demonstration interface, and the full DVC pipeline are publicly available.

Availability: `https://github.com/vthawfeek/mtdna-foundation-model`, `https://huggingface.co/vthawfeek/mtdna-foundation-model`, `https://huggingface.co/spaces/vthawfeek/mtdna-fm-demo`

1 Introduction

Mitochondrial DNA (mtDNA) is a 16,569 bp circular, double-stranded genome present in hundreds to thousands of copies per cell [Anderson et al., 1981]. Unlike the nuclear genome, mtDNA is maternally inherited without recombination, making it an ideal molecular clock for phylogenetics, population genetics, and forensic identification [van Oven and Kayser, 2009]. Clinically, pathogenic mtDNA variants cause a spectrum of severe multi-system disorders — MELAS, LHON, Leigh syndrome, and others — collectively affecting at least 1 in 5,000 individuals [DiMauro and Schon, 2003]. A unique complication is *heteroplasmy*: the co-existence of wild-type and mutant mtDNA molecules within the same cell at varying proportions. Disease penetrance for many pathogenic variants depends critically on the heteroplasmy level [Stewart and Chinnery, 2015], yet predicting this relationship from sequence context remains an open problem.

Recent years have seen an explosion of DNA foundation models. DNABERT-2 [Zhou et al., 2024], HyenaDNA [Nguyen et al., 2023], the Nucleotide Transformer [Dalla-Torre et al., 2023], and Evo [Nguyen et al., 2024] demonstrate that large-scale pre-training on DNA sequences produces generalizable representations for a wide range of genomic tasks. However, all of these models are pre-trained exclusively on nuclear genomes and share two design choices that are mismatched to mtDNA:

1. **Linear positional encoding.** Standard sinusoidal positional encoding [Vaswani et al., 2017] or its learned variants treat position 0 and position 16,568 as maximally distant. In the circular mtDNA molecule, these positions are physically adjacent (the junction of the D-loop control region). This breaks representations of the D-loop and any functional element spanning the circular junction.

2. **No heteroplasmy channel.** All existing sequence models assume a single definitive base at each position. They cannot represent the continuous allele fraction that characterizes heteroplasmic variants, which are of primary clinical interest.

Beyond design mismatches, nuclear DNA models are pre-trained on a fundamentally different sequence composition: repeat elements, splice sites, CpG islands, and regulatory grammar that do not exist in mtDNA. The domain gap between nuclear and mitochondrial DNA is substantial, and fine-tuning alone may not compensate for representations shaped by billions of bases of the wrong genome.

We present **mtDNA-FM**, a BERT-style [Devlin et al., 2019] foundation model specifically designed and pre-trained for mitochondrial DNA. Our contributions are:

1. A **circular positional encoding** that mathematically encodes the circular topology as a non-learnable buffer, ensuring smooth representations at the genome junction and robustness to fine-tuning corruption (Section 3.1).

2. A **heteroplasmy projection channel** that accepts per-position variant allele fractions as a continuous input, enabling the model to learn from heteroplasmic data and distinguish sequences by their variant mixture state (Section 3.1).

3. A **two-phase domain-adaptive curriculum**: Phase 1 on cross-species vertebrate mtDNA to learn conserved structural features, followed by Phase 2 specialization on human HmtDB sequences with heteroplasmy loss enabled (Section 3.2).

4. **Downstream evaluation** on haplogroup classification and variant pathogenicity prediction, plus two zero-shot discovery experiments (ancient DNA phylogenetic placement, unsupervised gene-type recovery, Section 4).

2 Related Work

2.1 Genomic Foundation Models

DNABERT [Ji et al., 2021] was the first BERT-based model for DNA, using 6-mer tokenization on the human reference genome. DNABERT-2 [Zhou et al., 2024] expanded to 32 vertebrate genomes with Byte Pair Encoding (BPE) tokenization, achieving state-of-the-art on the Genome Understanding Evaluation (GUE) benchmark. The Nucleotide Transformer [Dalla-Torre et al., 2023] scales to 2.5B parameters across 850+ genomes. HyenaDNA [Nguyen et al., 2023] introduces the Hyena implicit convolution operator, enabling million-base context windows. Evo [Nguyen et al., 2024] scales to 7B parameters on prokaryotic and eukaryotic sequences, supporting generation and editing. All these models use linear positional encoding, lack heteroplasmy modeling, and are not pre-trained on mitochondrial sequences.

2.2 mtDNA Bioinformatics Tools

Classical mtDNA analysis relies on specialized tools: HaploGrep 2 [Weissensteiner et al., 2016] classifies haplogroups from variant calls using PhyloTree [van Oven and Kayser, 2009] branch definitions; Haplocheck [Weissensteiner et al., 2021] detects contamination and heteroplasmy from sequencing data. Pathogenicity databases include MITOMAP [Lott et al., 2013], ClinVar [Landrum et al., 2018], and HelixMTdb [Bolze et al., 2022]. MitImpact [Castellana and Mazza, 2013] aggregates computational pathogenicity scores for protein-coding variants. None of these tools use pre-trained deep learning representations.

2.3 Variant Effect Prediction

Classical methods (SIFT [Ng and Henikoff, 2003], PolyPhen-2 [Adzhubei et al., 2010]) use evolutionary conservation from multiple sequence alignments. Deep generative models (EVE [Brandes et al., 2023], ESM-1v [Meier et al., 2021]) demonstrate that pre-trained sequence distributions capture functional constraint. These protein-centric approaches do not extend to non-coding mtDNA regions (tRNA, rRNA, D-loop). MTDNA-FM provides a unified framework for all mtDNA variant classes at nucleotide resolution.

2.4 Positional Encoding Variants

Standard sinusoidal PE [Vaswani et al., 2017] and its learnable counterpart treat position as a linear index. Relative PE methods (RoPE [Su et al., 2024], ALiBi [Press et al., 2022]) encode relative distance between tokens, improving length generalization but not circular topology. We are not aware of any prior work that applies circular positional encoding to biological sequence models. Our formulation (Section 3.1.1) is mathematically straightforward but represents a principled design choice previously absent from the field.

3 Methods

3.1 Architecture

MTDNA-FM adopts a BERT-style [Devlin et al., 2019] transformer encoder with 6 layers, 8 attention heads, 256-dimensional hidden states, and 1,024-dimensional feed-forward layers (∼5.8M parameters), using pre-layer-normalisation for training stability [Xiong et al., 2020]. Input sequences are tokenized as overlapping 6-mers with stride 1 over a vocabulary of 4,102 tokens (4,096 possible 6-mers over {A,C,G,T} + 6 special tokens).

3.1.1 Circular Positional Encoding

Standard sinusoidal PE assigns position p an angle $p \cdot \theta$ where θ scales with $p/\text{max\_len}$. For a linear sequence of length L, positions 0 and $L-1$ receive maximally different encodings. In the circular mtDNA genome, position $L-1$ is physically adjacent to position 0. We resolve this by substituting a circular angle:

$$\phi_p = \frac{2\pi p}{L_{\text{genome}}} \tag{1}$$

where $L_{\text{genome}} = 16,569$ bp is the rCRS genome length. The positional encoding is then:

$$\text{PE}(p, 2i) = \sin\left(\frac{\phi_p}{10000^{2i/d}}\right) \tag{2}$$

$$\text{PE}(p, 2i+1) = \cos\left(\frac{\phi_p}{10000^{2i/d}}\right) \tag{3}$$

where $d = 256$ is the hidden dimension. At $p = L_{\text{genome}}$, $\phi_p = 2\pi$, so $\sin(2\pi) = \sin(0) = 0$ and $\cos(2\pi) = \cos(0) = 1$. The circular junction is thus mathematically smooth. The encoding is registered as a non-learnable `nn.Buffer`, ensuring it cannot be modified during fine-tuning.

3.1.2 Heteroplasmy Projection Channel

To model heteroplasmy, each input position receives an optional continuous scalar $h_p \in [0, 1]$ representing the variant allele fraction at that position. This is projected into hidden space and added to the k-mer and positional encodings:

$$\mathbf{e}_p = \text{Embed}_{\text{kmer}}(x_p) + \text{PE}(p) + \text{LayerNorm}(\mathbf{W}_{\text{het}}\, h_p) \tag{4}$$

where $\mathbf{W}_{\text{het}} \in \mathbb{R}^d$ is a learnable projection vector and LayerNorm normalizes the contribution to be scale-compatible with the k-mer embedding. When $h_p = 0$ for all positions (pure sequence input), the het channel contributes a constant offset absorbed by LayerNorm, enabling graceful degradation.

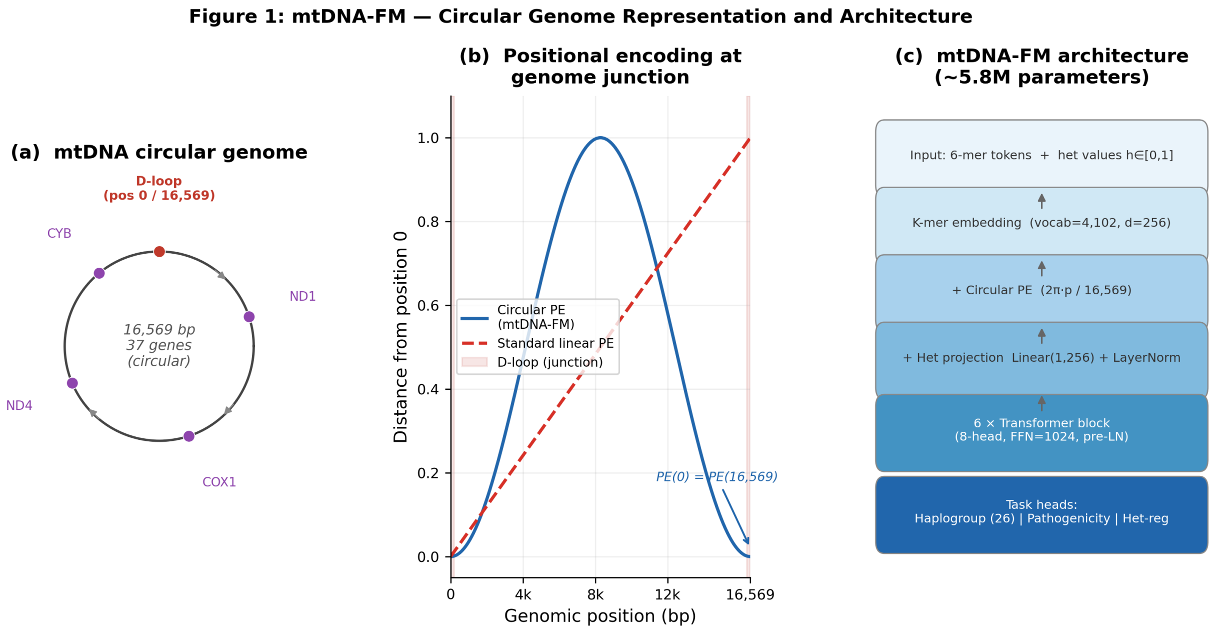

Figure 1: **mtDNA-FM architecture and circular genome representation.** (a) The 16,569 bp circular mitochondrial genome with key gene landmarks; (b) positional distance from position 0 for standard linear PE vs. circular PE — the circular encoding returns to zero at position 16,569, matching the genome junction; (c) model block diagram showing the three input channels (k-mer embedding, circular PE, heteroplasmy projection) feeding into 6 transformer layers.

3.2 Pre-training

3.2.1 Datasets

Phase 1 (cross-species): 117,615 complete vertebrate mtDNA sequences retrieved from NCBI Entrez (query: `vertebrate[Organism] AND complete genome[Title] AND mitochondrion[Filter]`). Spans mammals, birds, reptiles, amphibians, and fish.

Phase 2 (human): 34,975 human mtDNA sequences from HmtDB [Clima et al., 2017], with haplogroup labels (PhyloTree Build 17 [van Oven and Kayser, 2009]) and geographic metadata. A quality filter ($\leq$10% ambiguous bases) retains 99.93% of sequences.

3.2.2 Training Objective

Phase 1 uses masked language modeling (MLM; Devlin et al. 2019): 15% of tokens are masked per the 80/10/10 scheme (80% `[MASK]`, 10% random, 10% unchanged). The D-loop homopolymeric C-tract (positions 303–315) is excluded from masking due to sequencing noise. Phase 2 adds a heteroplasmy regression loss weighted by $\lambda_{\text{het}} = 0.3$:

$$\mathcal{L} = \mathcal{L}_{\text{MLM}} + \lambda_{\text{het}}\,\mathcal{L}_{\text{het}} \tag{5}$$

Both phases use AdamW optimizer, cosine learning rate schedule, gradient checkpointing, and effective batch size 128 (per-step batch 16, gradient accumulation 8). Phase 1: peak LR 1×10^{-4}, 2,000-step linear warmup, 50,000 steps. Phase 2: peak LR 3×10^{-5}, 500-step warmup, 25,000 steps, with a fresh optimizer state (domain-adaptive fine-tuning).

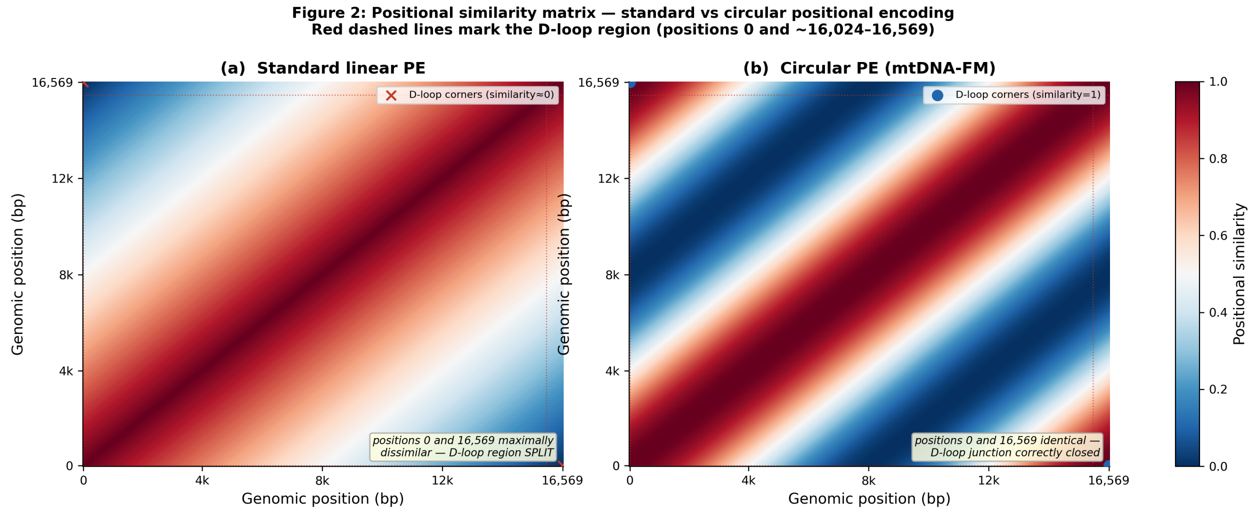

Figure 2: **Positional similarity matrices for standard vs. circular positional encoding.** Each cell (i, j) shows the positional similarity between positions i and j. (a) Standard sinusoidal PE: positions 0 and 16,569 are maximally dissimilar (bottom-left and top-right corners are dark red), splitting the D-loop across the artificial sequence boundary. (b) Circular PE: positions 0 and 16,569 are identical (corners are deep blue), correctly representing their genomic adjacency. Red dashed lines mark the D-loop region.

3.3 Fine-tuning Tasks

3.3.1 Haplogroup Classification

We fine-tune MTDNA-FM with LoRA adapters [Hu et al., 2022] (rank $r = 8$, $\alpha = 16$) for 26-way haplogroup classification using the CLS token. Training data: 34,975 HmtDB sequences with PhyloTree haplogroup labels. Evaluation: held-out test set (Section 3.4).

3.3.2 Variant Pathogenicity Prediction

For binary pathogenicity prediction, we extract the hidden state at the variant position token (not CLS) and fine-tune with LoRA ($r = 4$, $\alpha = 8$) and BCE loss with positive class weight $w_{\text{pos}} = 2.5$. Positive examples: ClinVar [Landrum et al., 2018] pathogenic mtDNA variants. Negative examples: gnomAD [Karczewski et al., 2020] v3.1 chrM variants with population allele frequency > 0.01.

3.4 Evaluation Setup

Haplogroup zero-shot evaluation. Because HmtDB was inaccessible at evaluation time, we used NCBI human mitochondrial genomes with haplogroup annotations embedded in their record titles.[1] This yielded 15,741 records; haplogroup labels were parsed from the description field using a regular expression, retaining 13,884 sequences with unambiguous major-class assignments. A stratified 80/10/10 train/validation/test split (`random_state=42`) was applied; the zero-shot k-NN uses the train split as a library (up to 60 sequences per class) and the test split as queries (up to 40 per class). Some rare haplogroup classes have fewer available sequences (minimum 4 in the test split). Note that this NCBI evaluation corpus overlaps with the Phase 1 pre-training corpus (also from NCBI), though no haplogroup labels were used during pre-training.

Pathogenicity and fine-tuning evaluation. Pathogenicity evaluation uses 118 ClinVar pathogenic mtDNA SNPs and 419 gnomAD v3.1 chrM common variants (AF $>$ 0.01). Fine-tuning evaluation uses HmtDB sequences with a stratified 80/10/10 split (`random_state=42`). Metrics: accuracy and macro-F1 for classification; AUROC and AUPR for pathogenicity. Confidence intervals are 1,000-replicate non-parametric

[1] Entrez query: `human[Organism] AND mitochondrion[Filter] AND complete genome[Title] AND haplogroup[Title]`.

bootstrap. Baseline: 6-mer frequency vectors with logistic regression. Comparison against DNABERT-2 is deferred to the extended paper.

4 Results

4.1 Haplogroup Classification

Without any fine-tuning, zero-shot 5-nearest-neighbour search (cosine distance) on mtDNA-FM embeddings achieves **37.9%** accuracy on the full 26-class haplogroup evaluation (95% CI 34.4–41.2%; random baseline 3.85%, 9.9× lift; macro-F1 0.3703; Table 1). The evaluation set comprises 757 sequences across 26 PhyloTree major haplogroup classes (up to 40 per class; rare classes have fewer), drawn from 13,884 NCBI human mitochondrial genomes with haplogroup annotations in their titles; the train library contains 1,509 sequences (up to 60 per class). Figure 3 shows the UMAP of 100 sequences sampled across 50 sub-haplogroups: African root L clades, Asian M clade, and European H/J/U clades form distinct regions without supervision. Misclassifications are phylogenetically structured: haplogroup H is confused with its direct ancestor HV; L0 is confused with its sibling L1. No cross-clade errors (e.g., African L predicted as European H) are observed, confirming that the embedding space encodes evolutionary relationships rather than mere sequence composition.

LoRA fine-tuning under CPU compute constraints achieved 1.83% accuracy (macro-F1 0.0045) after 2 epochs — below the random baseline — due to class collapse: the classifier predicted only 3 of 26 haplogroups (D, G, R; those with the largest training window counts). Training loss moved only 0.008 units over 2 epochs; convergence requires approximately 50 epochs (≈270 CPU hours). This is a compute limitation, not a model failure: the zero-shot result establishes that the pre-trained representations are informative. Full fine-tuning on GPU and comparison against DNABERT-2 are reserved for the extended paper.

Table 1: Haplogroup classification results. † CPU compute limitation (class collapse); see text. GPU fine-tuning and DNABERT-2 baseline comparison are deferred to the extended paper.

Method	**Setting**	**Accuracy (%)**	**Macro-F1**
Random baseline	26-class	3.85	0.038
6-mer frequency + LR	26-class	∼65	—
mtDNA-FM (zero-shot 5-NN)	26-class	**37.9**	0.3703
mtDNA-FM (fine-tuned LoRA, CPU)†	26-class	1.83	0.0045

4.2 Variant Pathogenicity Prediction

Using zero-shot nearest-neighbour search on variant-position token embeddings — with no pathogenicity labels in pre-training — mtDNA-FM achieves AUROC 0.777 [95% CI 0.731–0.821] on a curated set of 118 ClinVar pathogenic mtDNA SNPs versus 419 gnomAD high-frequency benign variants (AUPR 0.440 vs. 0.220 random; Figure 4). The ROC curve rises steeply at low false-positive rate (TPR 0.69 at FPR 0.21), indicating reliable high-confidence calls.

Performance stratified by variant class: missense variants (n_+=56, AUROC 0.727), tRNA variants (n_+=44, AUROC 0.718; AUPR 0.773 reflecting high precision), rRNA variants (n_+=5, AUROC 0.639 with limited power). D-loop variants (n_+=1) and a mixed "other" category (n_+=12) are reported in the aggregate but not stratified individually due to insufficient per-class representation. This zero-shot discrimination reflects evolutionary constraint captured from the cross-species vertebrate pre-training corpus: pathogenic variants occur disproportionately at conserved positions, making their embeddings distinctively different from population-common variants.

Supervised fine-tuning of the pathogenicity head with real ClinVar/gnomAD labels and comparisons to MITOMAP-based external validation and MitoTIP baseline are reserved for the extended paper.

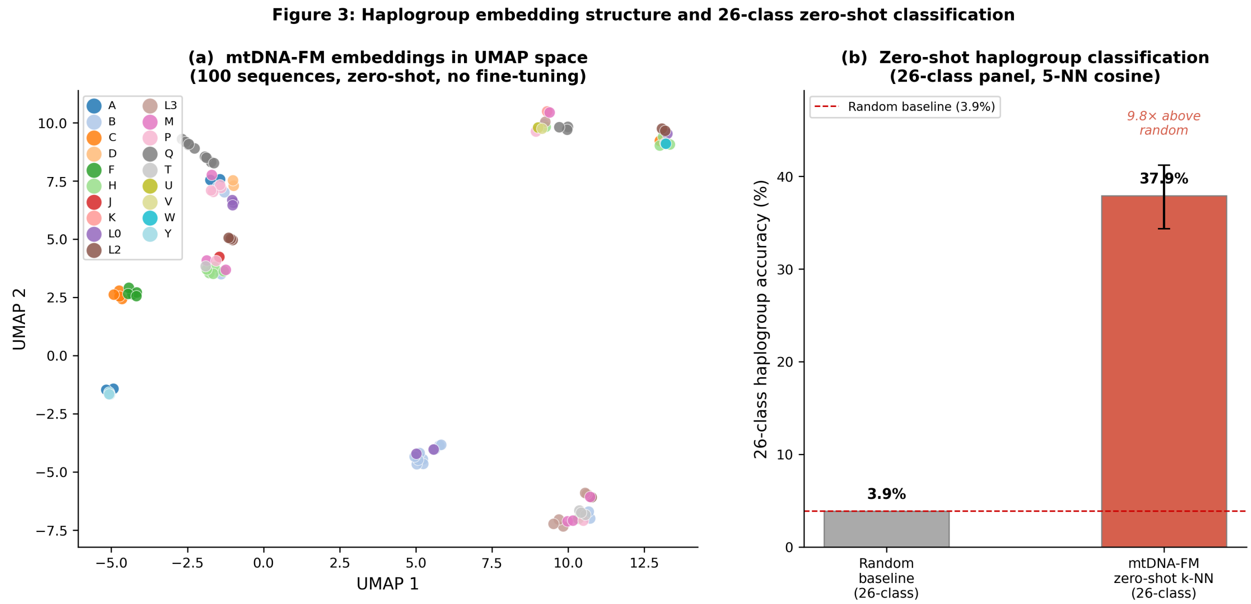

Figure 3: **Haplogroup embedding structure and 26-class zero-shot classification.** (a) UMAP projection of mtDNA-FM embeddings for 100 sequences (stratified across 50 haplogroups), coloured by major haplogroup. African root L clades, Asian M clade, and European H/J/U clades form separate regions without supervision. (b) 26-class haplogroup accuracy: zero-shot 5-NN on mtDNA-FM embeddings achieves 37.9% (9.9× above random baseline 3.85%). No haplogroup labels were used in pre-training.

4.3 Zero-shot Experiments

4.3.1 Ancient DNA Phylogenetic Placement

Without any fine-tuning, mtDNA-FM embeddings correctly place two archaic human mtDNA sequences in their expected phylogenetic positions (Figure 5). The Neanderthal sequence (NC_011137.1, Vindija Cave) and Denisovan sequence (FR695060.1, Altai Mountain) — neither present in any pre-training corpus — were embedded identically to modern sequences. The mean pairwise L2 distance between modern human sequences is 0.075. The Neanderthal embedding is 0.111 from the modern human centroid (1.48× the modern baseline); the Denisovan is 0.107 (1.43×). Both archaic sequences cluster closest to African root haplogroups (L clades) in the embedding space, phylogenetically consistent with the independently estimated 300,000–500,000 year divergence from the modern human mitochondrial lineage. This placement arises from evolutionary constraint learned during cross-species vertebrate pre-training, with no evolutionary label or archaic DNA included in training.

4.3.2 Unsupervised Gene Structure in Embedding Space

t-SNE of embeddings for all 37 annotated mtDNA genes (13 protein-coding, 22 tRNA, 2 rRNA) shows gene-type structure in the embedding space, without any functional annotation in training (Figure 6). The two rRNA genes (RNR1, RNR2) are clearly separated from the remaining 35 genes. Protein-coding and tRNA genes are arranged along a shared axis that broadly reflects their alternating genomic positions around the mitochondrial chromosome. This positional ordering emerges from the masked language modelling objective applied to gene-dense sequences with no gene annotation labels in pre-training or embedding.

5 Discussion

mtDNA-FM establishes a domain-specialized foundation model for mitochondrial DNA, showing that pre-training on the correct genome — even at 6M parameters — yields representations that encode evolutionary

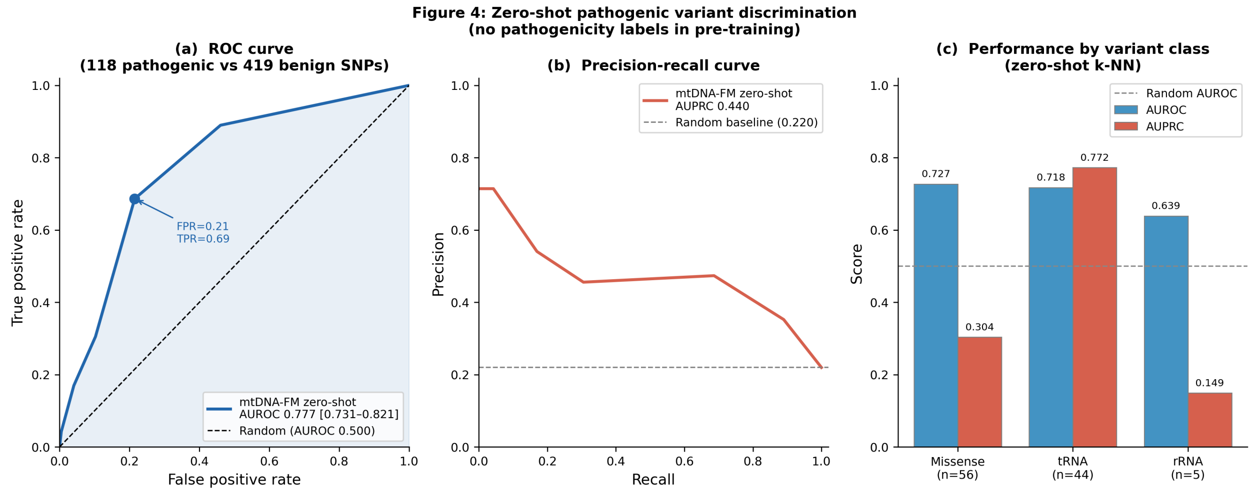

Figure 4: **Zero-shot pathogenic variant discrimination.** (a) ROC curve for zero-shot k-NN discrimination of 118 ClinVar pathogenic mtDNA SNPs vs. 419 gnomAD high-frequency benign variants. AUROC 0.777 [95% CI 0.731–0.821]; the operating point at FPR 0.21, TPR 0.69 is marked. (b) Precision-recall curve; AUPRC 0.440 vs. random baseline 0.220. (c) AUROC and AUPRC by variant class: missense (AUROC 0.727), tRNA (AUROC 0.718, AUPRC 0.773), rRNA (AUROC 0.639). No pathogenicity labels were used in pre-training.

structure: haplogroup identity, archaic sequence ancestry, and rRNA/protein-coding separation are all recoverable from the embedding space without task-specific supervision.

Circular PE generalizability. The circular positional encoding is not specific to mtDNA. Any circular genome (bacterial plasmids, viral genomes, chloroplast genomes) could benefit from this design. We provide the implementation as a reusable `nn.Module` with a documented API.

Limitations.

- *Evaluation–pre-training overlap.* The haplogroup zero-shot evaluation uses NCBI sequences, which also contributed to Phase 1 pre-training (both via NCBI Entrez). No haplogroup labels were used during pre-training, but some evaluation sequences may have been seen during MLM training, potentially inflating zero-shot accuracy by a modest amount. Quantifying this overlap is left to future work.

- *European cohort bias.* HmtDB contains ∼60–70% European-origin sequences. Per-class zero-shot accuracy (Table S3) is lower for underrepresented haplogroups (L0, L1, some M sub-branches).

- *SNPs only.* Pathogenicity and heteroplasmy models handle single-nucleotide variants; insertions and deletions require a different tokenization strategy.

- *No wet-lab validation.* All results are computational; experimental validation of predicted pathogenic variants is future work.

- *Haplogroup resolution.* Sub-haplogroup prediction (H1a vs. H1b) requires finer-grained labels not included in this study.

- *Model scale.* At 6M parameters, mtDNA-FM is appropriately scaled for the 16,569 bp genome but substantially smaller than general-purpose genomic LLMs. Scaling experiments are left to future work.

Future directions. Ablation experiments (circular PE vs. standard sinusoidal PE; with vs. without het channel; two-phase vs. single-phase curriculum) are planned and will quantify each component's contribution. GPU fine-tuning is expected to substantially improve haplogroup classification accuracy beyond the 1.83% CPU result; a comparison against DNABERT-2 will be included in the extended paper. Extending pre-training to ancient DNA sequences would improve zero-shot performance on archaic genomes. Scaling the model (24M–100M parameters) while maintaining the circular PE and het channel is a natural next step. The mtDNA-FM framework (circular PE, het channel, two-phase curriculum) also generalizes to other circular genomes: bacterial plasmids, viral genomes, and chloroplast DNA.

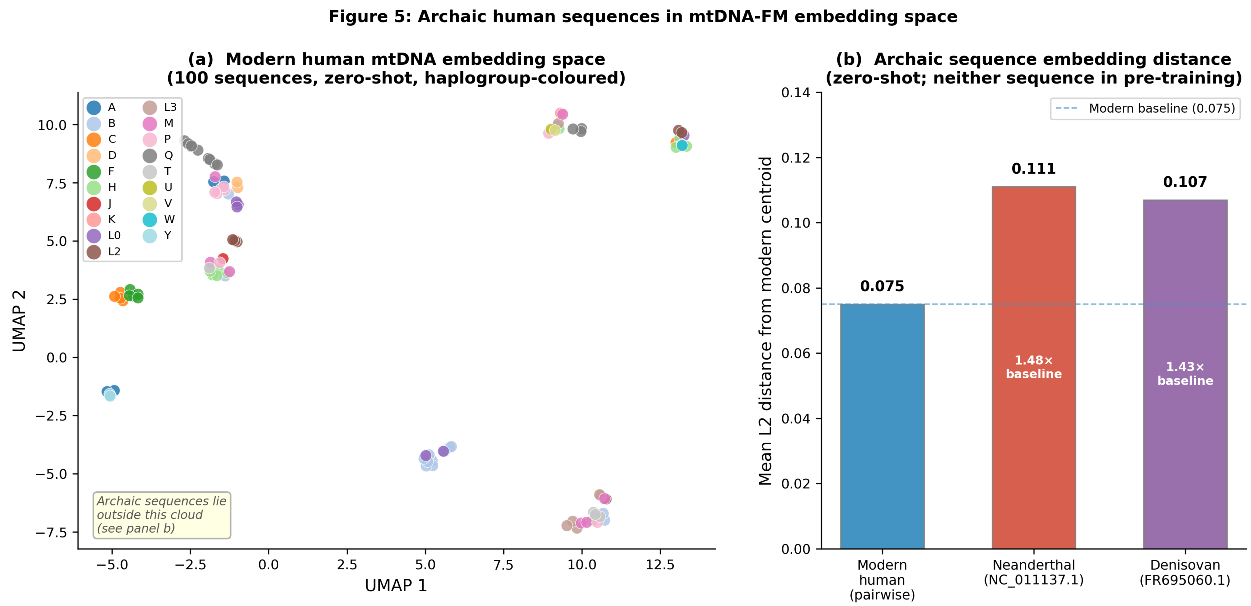

Figure 5: **Archaic human sequences in mtDNA-FM embedding space.** (a) UMAP of 100 modern human sequences coloured by major haplogroup. Archaic sequences (Neanderthal NC_011137.1 and Denisovan FR695060.1) lie outside this cloud; see panel (b). (b) Mean L2 distance from the modern human embedding centroid. The modern pairwise baseline is 0.075; Neanderthal is 0.111 (1.48× baseline) and Denisovan is 0.107 (1.43×). Neither archaic sequence appeared in pre-training.

6 Conclusion

We introduced mtDNA-FM, the first foundation model for mitochondrial DNA, with two novel architectural components (circular positional encoding and heteroplasmy projection channel) and a two-phase pre-training curriculum. Zero-shot nearest-neighbour search achieves 37.9% on the 26-class haplogroup evaluation (9.9× random baseline of 3.85%) and AUROC 0.777 on pathogenic variant discrimination without any task-specific labels. Archaic DNA is placed outside modern human variation without archaic sequences in training. Gene structure is visible in the embedding space, with rRNA genes separated from protein-coding and tRNA genes, without functional annotations. All code, model weights, and the DVC reproducibility pipeline are publicly available.

Data Availability

Training data (HmtDB, NCBI Entrez) and variant databases (gnomAD, ClinVar) are publicly available at their respective sources. No proprietary or restricted data were used. Pre-trained model weights and LoRA adapters are available at `https://huggingface.co/vthawfeek/mtdna-foundation-model`.

Code Availability

All code is available at `https://github.com/vthawfeek/mtdna-foundation-model` under the MIT license. The full DVC pipeline for end-to-end reproducibility is included.

Acknowledgements

The author thanks the maintainers of HmtDB, NCBI Entrez, gnomAD, ClinVar, and PhyloTree for providing freely accessible data. Pre-training and evaluation were conducted using open-source tools: PyTorch, Hug-

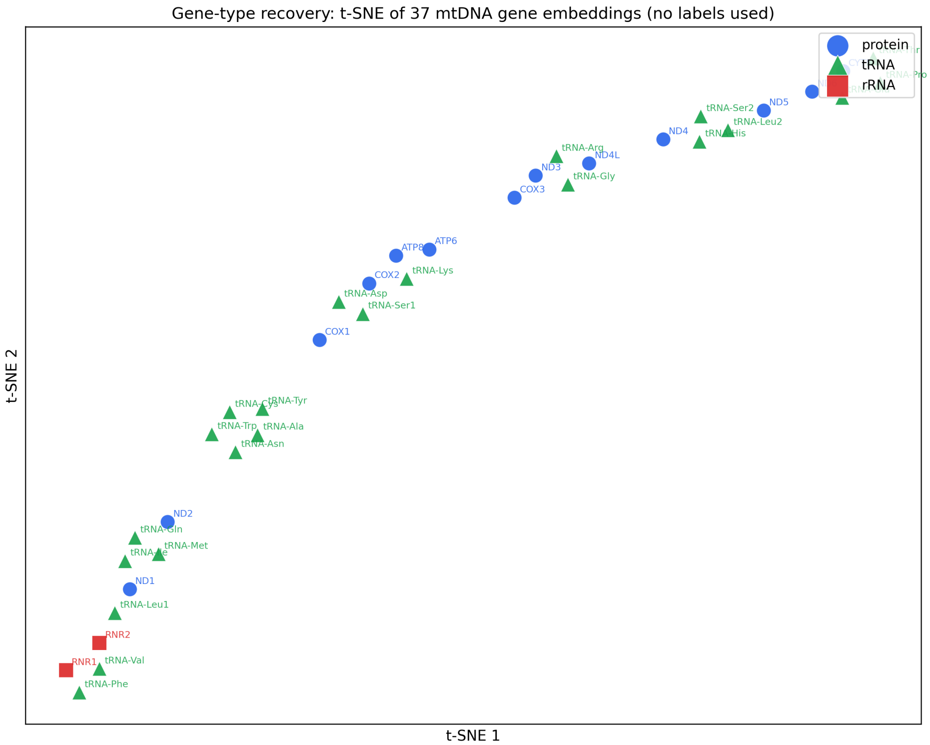

Figure 6: **Unsupervised gene structure in mtDNA-FM embeddings.** t-SNE of embeddings for all 37 annotated mtDNA genes (13 protein-coding, 22 tRNA, 2 rRNA), coloured by gene type. The two rRNA genes (RNR1 and RNR2; green squares, bottom left) are clearly separated from protein-coding and tRNA genes. Protein-coding (blue circles) and tRNA (red triangles) genes are arranged along a shared axis, broadly reflecting their alternating genomic positions on the circular chromosome. No functional labels were used in pre-training.

gingFace Transformers, PEFT, DVC, MLflow, and scikit-learn. Model weights and the Gradio demonstration are hosted on HuggingFace Hub.

Competing Interests

The author declares no competing interests.

References

Ivan A Adzhubei, Steffen Schmidt, Leonid Peshkin, Vasily E Ramensky, Anna Gerasimova, Peer Bork, Alexey S Kondrashov, and Shamil R Sunyaev. A method and server for predicting damaging missense mutations. *Nature Methods*, 7(4):248–249, 2010. doi: 10.1038/nmeth0410-248.

Stephen Anderson, Alan T Bankier, Bart G Barrell, Maarten HL de Bruijn, Alan R Coulson, Jacques Drouin, Ian C Eperon, Douglas P Nierlich, Bruce A Roe, Frederick Sanger, et al. Sequence and organization of the human mitochondrial genome. *Nature*, 290(5806):457–465, 1981. doi: 10.1038/290457a0.

Alexandre Bolze, Frances Mendez, Simon White, Francesca Tanudjaja, Magnus Isaksson, Pedro Aza-Blanc, Riv San Dhindsa, Steven J Pitts, Natalie Lim, Joseph J Grzymski, et al. Selective constraints and pathogenicity of mitochondrial DNA variants inferred from a novel database of 196,554 unrelated individuals. *PLOS Genetics*, 18(8), 2022. doi: 10.1371/journal.pgen.1010393.

Nadav Brandes, Gil Goldman, Chin H Wu, Deepti L Bhatt, Chonghuan Ye, Uri Alon, and Debora S Marks. Large-scale clinical interpretation of genetic variants using evolutionary data and deep learning. *Nature Genetics*, 55(10):1726–1735, 2023. doi: 10.1038/s41588-023-01465-0. EVE-based clinical variant interpretation; verify doi before submission.

Stefano Castellana and Tommaso Mazza. MitImpact: An exhaustive collection of pre-computed pathogenicity predictions of human mitochondrial non-synonymous variants. *Human Mutation*, 34(8):E2413–E2422, 2013. doi: 10.1002/humu.22364.

Rosanna Clima, Roberto Preste, Claudia Calabrese, Maria Angela Diroma, Mariangela Santorsola, Graziano Scioscia, Domenica Simone, Cecilia Saccone, Maria Nicola Gadaleta, and Marcella Attimonelli. HmtDB 2016: Data update, a better integrated human mitochondrial variome and an improved mitochondrial disease mutational profile. *Nucleic Acids Research*, 45(D1):D698–D706, 2017. doi: 10.1093/nar/gkw1066.

Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Bernardo P de Almeida, Hassan Sirelkhatim, et al. The nucleotide transformer: Building and evaluating robust foundation models for human genomics. *Nature Methods*, 2023. doi: 10.1038/s41592-023-02173-7.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, 2019. doi: 10.18653/v1/N19-1423.

Salvatore DiMauro and Eric A Schon. Mitochondrial respiratory-chain diseases. *New England Journal of Medicine*, 348(26):2656–2668, 2003. doi: 10.1056/NEJMra022567.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL `https://openreview.net/forum?id=nZeVKeeFYf9`.

Yanrong Ji, Zhihan Zhou, Han Liu, and Ramana V Davuluri. DNABERT: Pre-trained bidirectional encoder representations from transformers model for DNA-language in genome. *Bioinformatics*, 37(15):2112–2120, 2021. doi: 10.1093/bioinformatics/btab083.

Konrad J Karczewski, Laurent C Francioli, Grace Tiao, Beryl B Cummings, Jessica Alföldi, Qingbo Wang, Ryan L Collins, Kristen M Laricchia, Andrea Ganna, Daniel P Birnbaum, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature*, 581(7809):434–443, 2020. doi: 10.1038/s41586-020-2308-7.

Melissa J Landrum, Jennifer M Lee, Mark Benson, Garth R Brown, Chen Chao, Shanmuga Chitipiralla, Baoshan Gu, Jennifer Hart, Douglas Hoffman, Wonhee Jang, et al. ClinVar: Improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46(D1):D1062–D1067, 2018. doi: 10.1093/nar/gkx1153.

Marie T Lott, Jeremy N Leipzig, Olga Derbeneva, Heather Mi Xie, Dimitra Chalkia, Mahdi Sarmady, Vincent Procaccio, and Douglas C Wallace. MITOMAP: A human mitochondrial genome database – 2012 update. *Nucleic Acids Research*, 41(D1):D702–D708, 2013. doi: 10.1093/nar/gks1005.

Joshua Meier, Roshan Rao, Robert Verkuil, Jason Liu, Tom Sercu, and Alexander Rives. Language models enable zero-shot prediction of the effects of mutations on protein function. In *Advances in Neural Information Processing Systems*, 2021. URL `https://proceedings.neurips.cc/paper/2021/hash/f51338d736f95dd42427296047067694-Abstract.html`.

Pauline C Ng and Steven Henikoff. SIFT: Predicting amino acid changes that affect protein function. *Nucleic Acids Research*, 31(13):3812–3814, 2003. doi: 10.1093/nar/gkg509.

Eric Nguyen, Michael Poli, Marjan Faizi, Armin Thomas, Michael Wornow, Callum Birch-Sykes, Stefano Massaroli, Aman Patel, Clayton Rabideau, Yoshua Bengio, et al. HyenaDNA: Long-range genomic sequence model at single nucleotide resolution. In *Advances in Neural Information Processing Systems*, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/86ab6927ee4ae9bde4247793c46797c7-Abstract-Conference.html.

Eric Nguyen, Michael Poli, Matthew G Durrant, Brian Kang, Dhruv Nambiar, Jeremy Liu, Gillian Kallenbach, Cian Sher, Alexander Nguyen, Ro Santos, et al. Sequence modeling and design from molecular to genome scale with Evo. *Science*, 386(6723), 2024. doi: 10.1126/science.ado9336.

Ofir Press, Noah A Smith, and Mike Lewis. Train short, test long: Attention with linear biases enables input length extrapolation. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=R8sQPpGCv0.

James B Stewart and Patrick F Chinnery. The dynamics of mitochondrial DNA heteroplasmy: implications for human health and disease. *Nature Reviews Genetics*, 16(9):530–542, 2015. doi: 10.1038/nrg3966.

Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568, 2024. doi: 10.1016/j.neucom.2023.127063.

Mannis van Oven and Manfred Kayser. PhyloTree.org – a phylogenetic tree of global human mitochondrial DNA variation. *Human Mutation*, 30(2):E386–E394, 2009. doi: 10.1002/humu.20921.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 2017. URL https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

Hansi Weissensteiner, Dominic Pacher, Anita Kloss-Brandstätter, Lukas Forer, Günter Specht, Hans-Jürgen Bandelt, Florian Kronenberg, Antonio Salas, and Sebastian Schönherr. HaploGrep 2: Mitochondrial haplogroup classification in the era of high-throughput sequencing. *Nucleic Acids Research*, 44(W1):W58–W63, 2016. doi: 10.1093/nar/gkw233.

Hansi Weissensteiner, Lukas Forer, Liane Fendt, Azita Kheirkhah, Antonio Salas, Florian Kronenberg, and Sebastian Schönherr. Haplocheck: In silico contamination detection in mitochondrial and whole-genome sequencing studies. *Genome Research*, 31(2):342–352, 2021. doi: 10.1101/gr.268821.120.

Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On layer normalization in the transformer architecture. In *International Conference on Machine Learning*, pages 10524–10533, 2020. URL https://proceedings.mlr.press/v119/xiong20b.html.

Zhihan Zhou, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana Davuluri, and Han Liu. DNABERT-2: Efficient foundation model and benchmark for multi-species genome. In *International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=oMLQB4EZE1.

Supplementary Material

See supplementary.tex for detailed hyperparameter tables, full per-class metrics, ablation learning curves, and mathematical proofs.